

7	(a) (i) minimum photon energy minimum energy to remove an electron (from the surface)	B1 B1 [2]
	(ii) <i>either</i> maximum KE is photon energy – work function energy <i>or</i> max KE when electron ejected from the surface energies lower than max because energy required to bring electron to the surface	B1 B1 [2]
	(b) (i) threshold frequency = $1.0 \times 10^{15} \text{ Hz}$ (allow $\pm 0.05 \times 10^{15}$) work function energy = hf_0 $= 6.63 \times 10^{-34} \times 1.0 \times 10^{15}$ $= 6.63 \times 10^{-19} \text{ J}$ (allow alternative approaches based on use of co-ordinates of points on the line)	C1 C1 A1 [3]
	(ii) sketch: straight line with same gradient displaced to right	M1 A1 [2]
	(iii) intensity determines number of photons arriving per unit time intensity determines number of electrons per unit time (not energy)	B1 B1 [2]